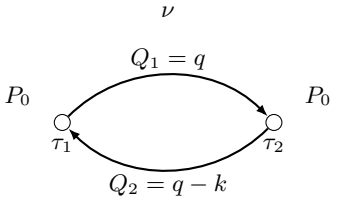
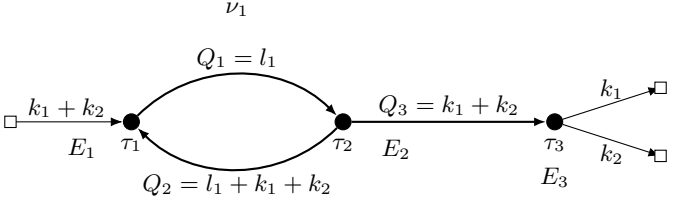

ν_1
$Q_1 = l_3$
kE
$Q_2 = k_{321}$
kE
τ_1
τ_2
$Q_3 = l_3 - k_{321} - k_L$

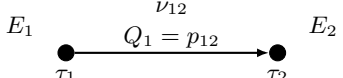
(a)


ν
$Q_1 = q$
P_0
P_0
τ_1
τ_2
$Q_2 = q - k$
ν

(b)


ν_1
$Q_1 = l_1$
$k_1 + k_2$
$Q_3 = k_1 + k_2$
k_1
E_1
τ_1
τ_2
E_2
τ_3
k_2
E_3
$Q_2 = l_1 + k_1 + k_2$

(c)


ν_{12}
E_1
$Q_1 = p_{12}$
E_2
τ_1
τ_2

(d)